CHAPTER 1 : What are Computers?

Question 1: 10 examples from daily life where 3D printers are making an impact

Custom Orthopedic Casts & Splints: Replacing bulky, itchy plaster casts with lightweight, breathable, waterproof 3D-printed lattices tailored precisely to a patient's limb scan.

Clear Dental Aligners & Retainers: Orthodontists take a 3D intraoral scan of your teeth and print accurate resin molds to shape clear aligners (like Invisalign).

Eyewear Frames: Brands produce custom, ultra-lightweight glasses frames printed to match the exact dimensions of a person's bridge and temples.

Custom Running Shoe Midsoles: Footwear companies (like Adidas 4D) print ~~tratice~~ lattice midsoles engineered with poned cushioning calibrated to stride impact.

Architectural House Construction: Large gantry 3D printers extrude specialiped quick-setting concrete layer-by-layer to construct ~~doraable~~ durable, low-cost home walls within ~~digs~~ days.

Prosthetic Limbs for Children: Open-source initiatives print affordable, mechanically functional prosthetic hands and arms that can be easily reprinted as a child grows.

Aircraft & Automotive Lightweight Brackets: Aerospace and automotive engineers print complex, organic metal geometries (generative design) that reduce component weight without sacrificing strength.

On-Demand Replacement Home Parts: Printing discontinued plastic knobs, gears, vacuum brackets, or dishwasher wheels instead of discarding entire household appliances.

Film Props and Practical Effects: Film studios print complex armor, helmets, movie props, and miniature terrain directly from 3D digital sculpts.

Culinary Art & Intricate Confectionery: Food printers deposit tempered chocolate, sugar paste, or pasta dough into geometric designs impossible to craft by hand.

Question 2: Difference between machines that can be programmed and machines that cannot be programmed

Non-programmable machines: Built with a fixed mechanical or electronic design to perform exactly one task. Their logic is hardwired into their physical structure. For instance, a basic mechanical hand-drill, a pendulum clock, or a classic bicycle cannot change its function; to make it do something else, you would have to physically rebuild it.

Programmable machines: General-purpose hardware that separates the machine itself from the instructions it follows. It includes an execution engine (like a CPU) and memory. By changing the software stored in memory, the exact same machine can compute financial alpha models, play an

HD video, or compile code without altering a single physical wire.

Question 3: Your childhood dream, in a world of computers
(Example: Dreaming of becoming an astronomer / space explorer)

(a) How computers shape this dream today: Modern astronomy is essentially big data computation. Massive telescopes do not rely on humans staring through an eyepiece; instead, digital sensor arrays collect petabytes of sky surveys. Supercomputers run gravitational wave simulations, clean atmospheric distortion in real time using adaptive optics, and apply machine learning models to detect exoplanet transits.

(b) How my B.Tech journey connects: Developing strong foundations in algorithms, linear algebra, and AI models enables building automated computer vision pipelines to analype telescope imagery, track satellite constellations, or optimipe autonomous rover navigation.

(c) Reflection: Exploring this makes the dream feel far more grounded and actionable. Instead of passive wonder, it turns into an active engineering challenge that can be contributed to directly through software and data science.